EXP. NO. : 2
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Study of different types of network cables and crimping of cables with RJ45 connector

AIM :

To study the different types of network cables.

a). understanding different types of network cables. The different types of cables used in networking are.

1. Unshielded twisted pair (UTP) cable.
2. Shielded twisted pair (STP) cable.
3. Coaxial cable
4. Fibre optic cable

Cable type	category	Maximum data transmission	Advantages / Disadvantages	Applications
UTP	category 3 category 5 category 5e	10 bps upto 100 Mbps 1 Gbps	Advantages: → cheaper. → Easy to install Disadvantages: → More prone to EMI & Noise	10 Base - T Ethernet Fast Ethernet
STP	category 6. 6a	10 Gbps	Advantages: → Shielded → Faster than UTP → Less to noise	Gigabit - Ethernet

coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: → High band width → versatile Disadvantage: → Limited distance → Cost → size, in bulky	Speed of signal is 500 m- Television network
Fibre optic cable	single mode, multimedia	100 Gbps	Advantage: → High speed → High band width → High security Disadvantage: → Expensive → Requires skilled Installers	Maximum distance of fibre optic cable is 100 metres

RESULT:

Thus, the different types of network Cables were studied and verified.